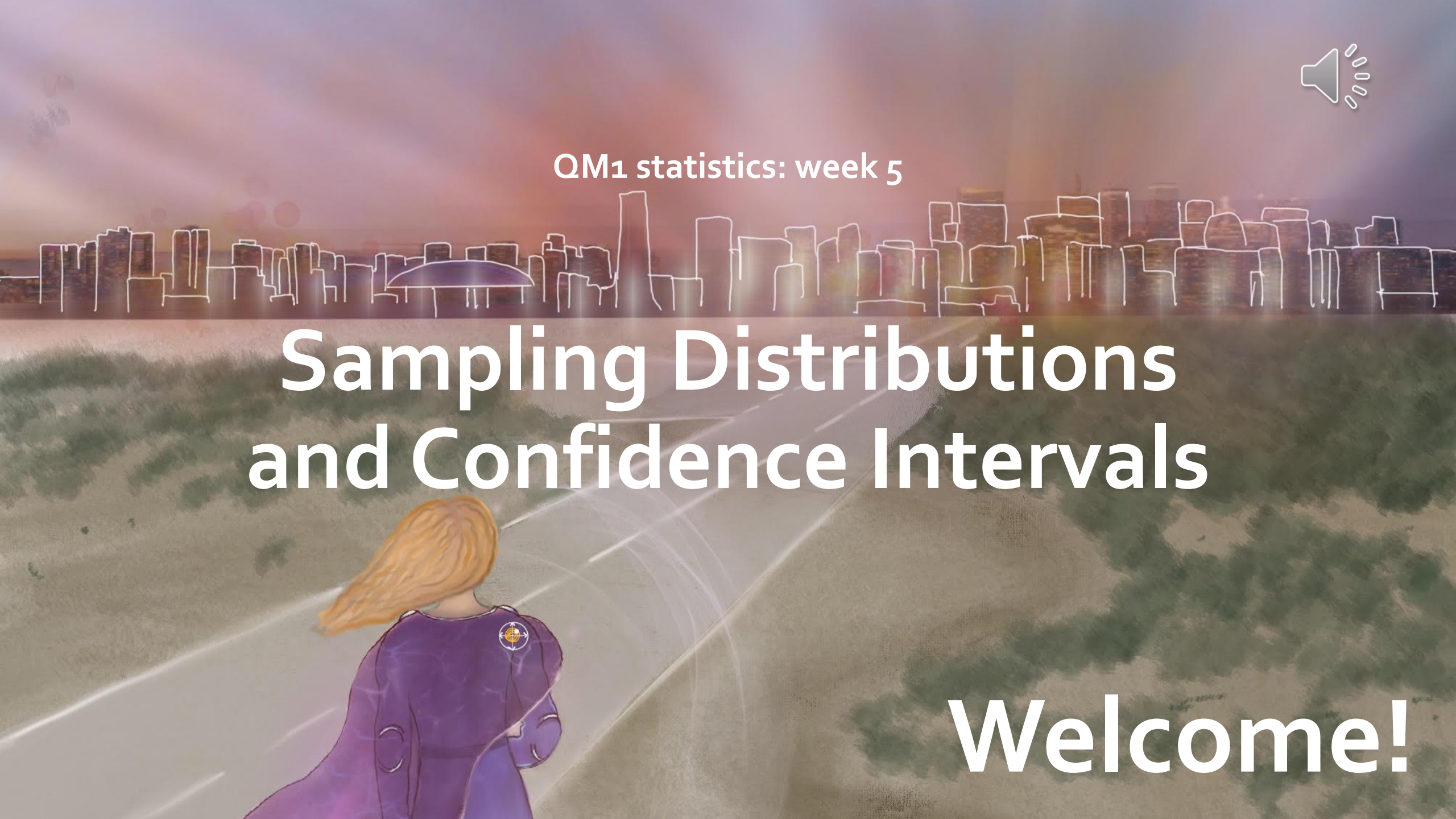
QM1 statistics: week 5

Sampling Distributions and Confidence Intervals

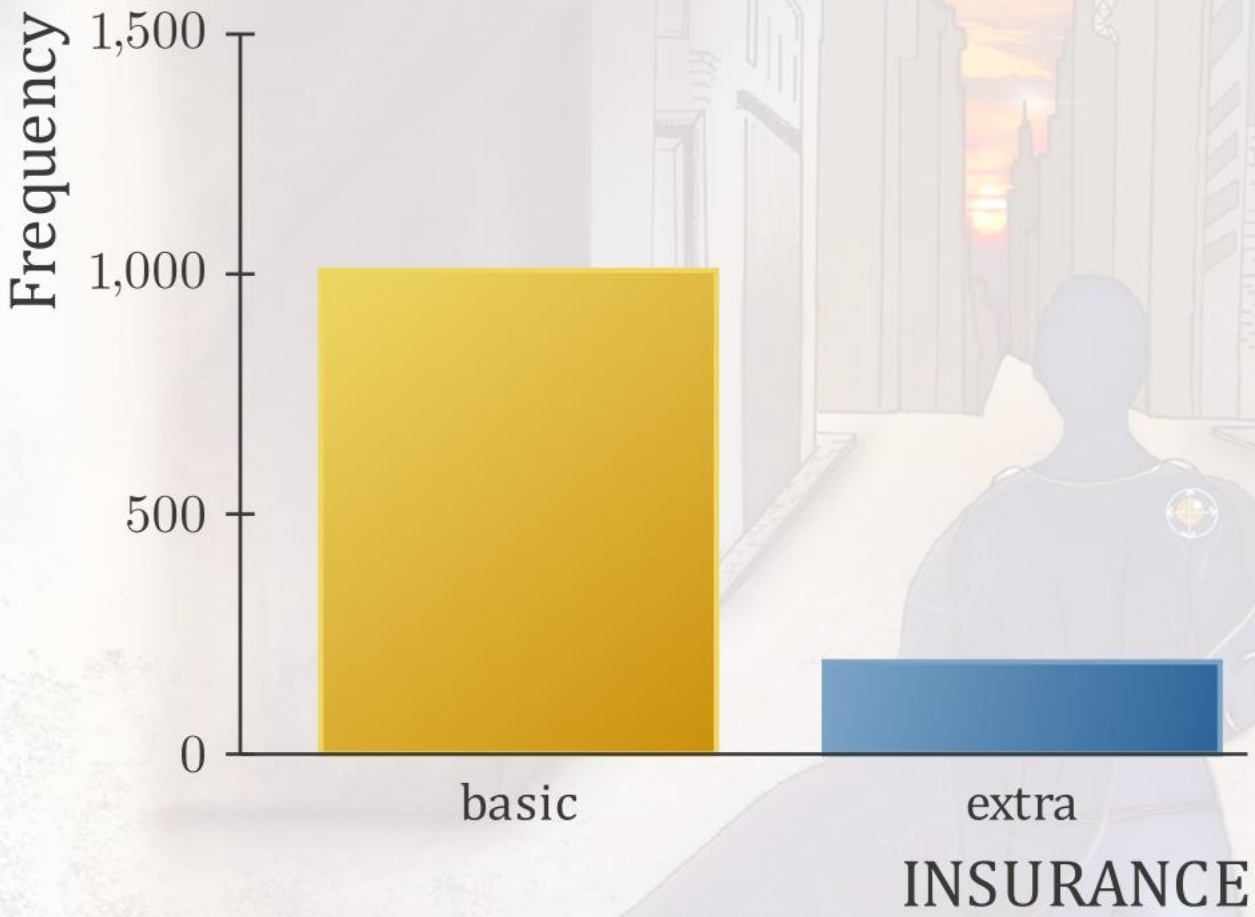


Welcome!

Risking your health

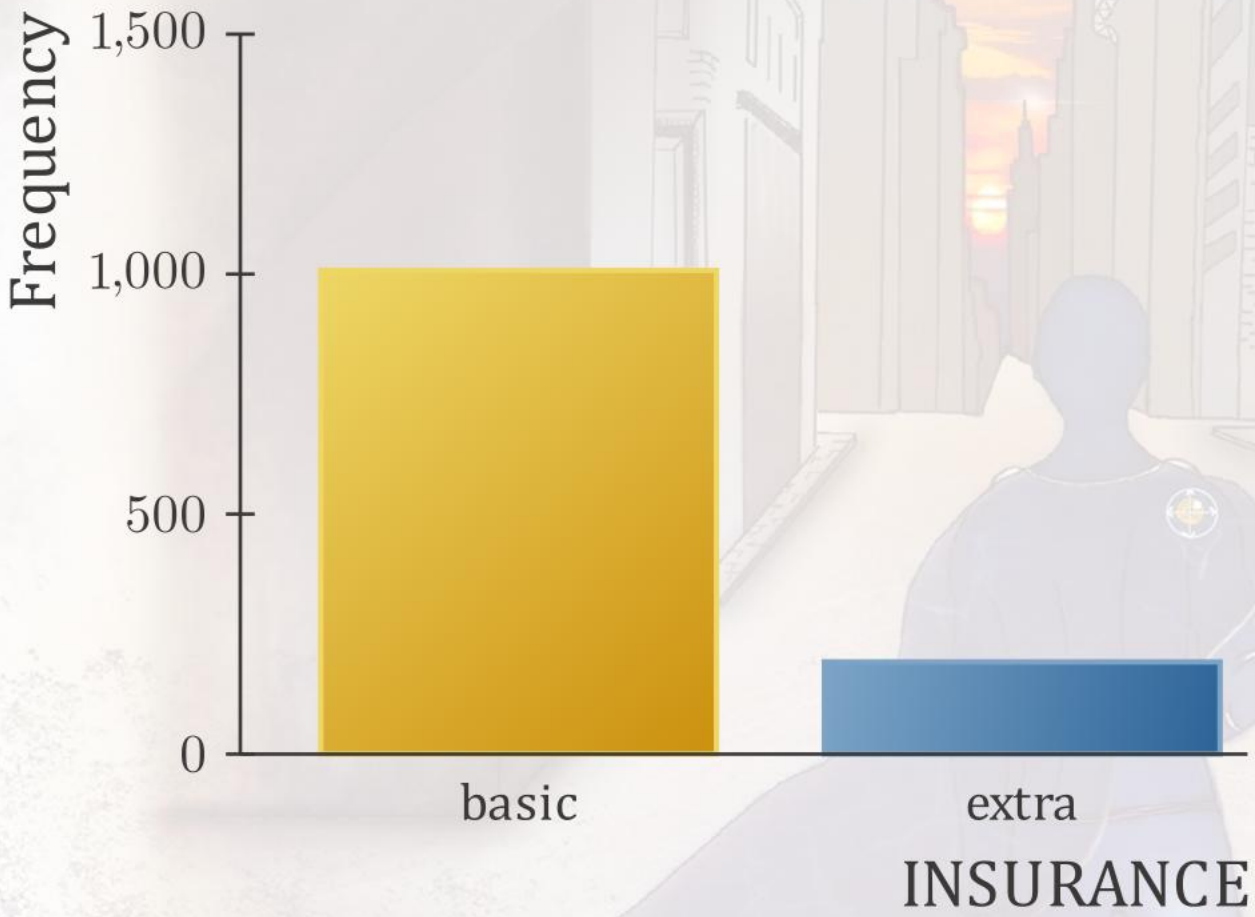
Sampling distributions

Being a superhero is a dangerous profession



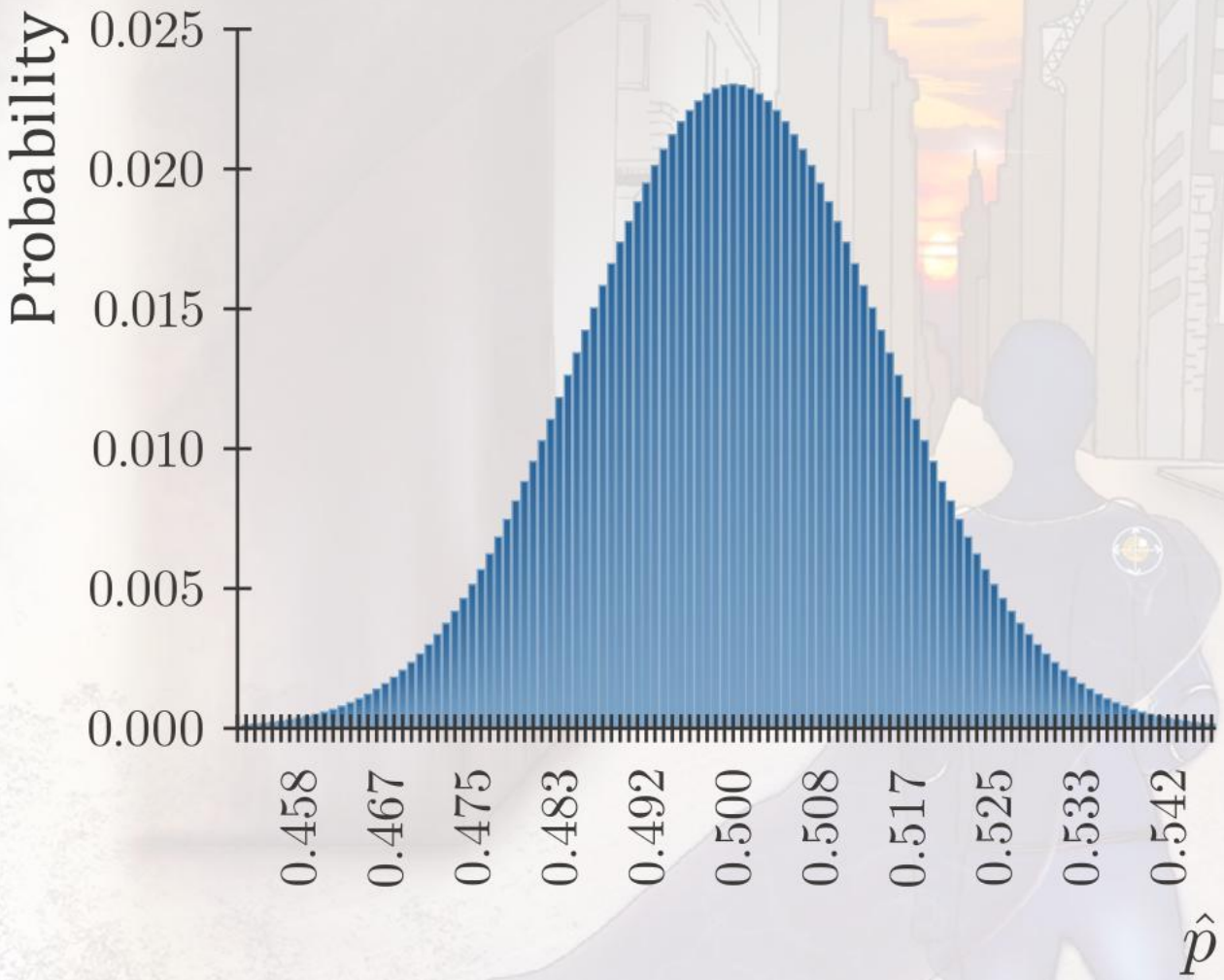
- In 2000, Professor Xavier conducted a survey. It asked a (simple random) sample of *n* = 1200 heroes whether they had supplementary health insurance
- *x* = 192 respondents said they did. Which amounts to only 16%
  - $\% = \frac{x}{n} * 100 = \frac{192}{1200} * 100 = 16$
- We can also express this percentage as a **proportion**:
  - $\hat{p} = \frac{x}{n} = \frac{192}{1200} = 0.16$
- Same information, but now expressed in a value between 0 and 1
- Why the hat (^)? Because $\hat{p}$ is just an **estimate**

Populations versus samples



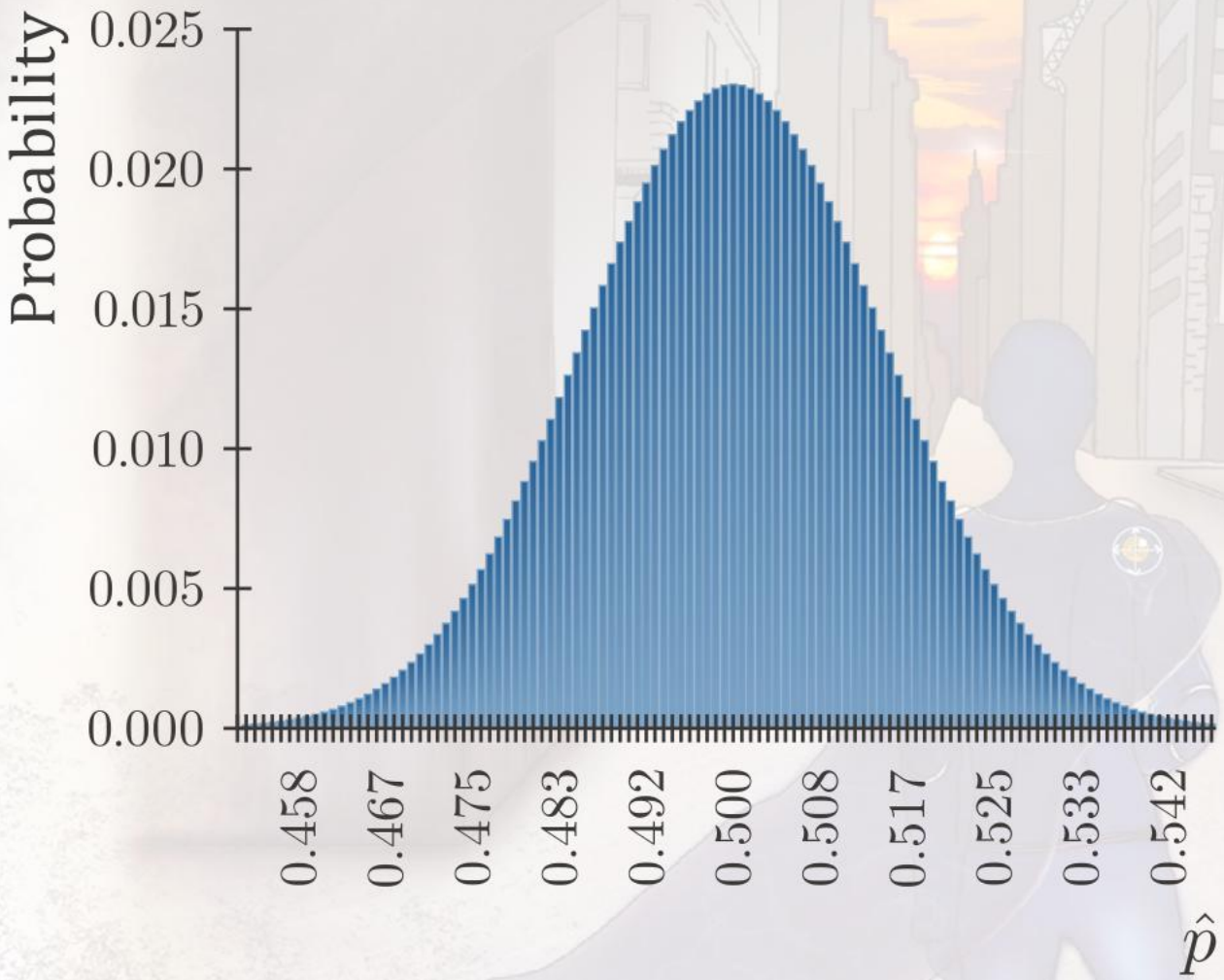
- We're actually interested in <u>all</u> superheroes (the population)
- The n = 1200 individuals are just a random sample, so $\hat{p}$ is not necessarily accurate
- What we really want to know is ***p***: the **population proportion** of heroes with proper insurance. We call this a population **parameter**
- $\hat{p}$ is just a **sample statistic** – in other words, an **estimate** of the population proportion
- So how can we tell if it's any good?

Statistics will vary from sample to sample



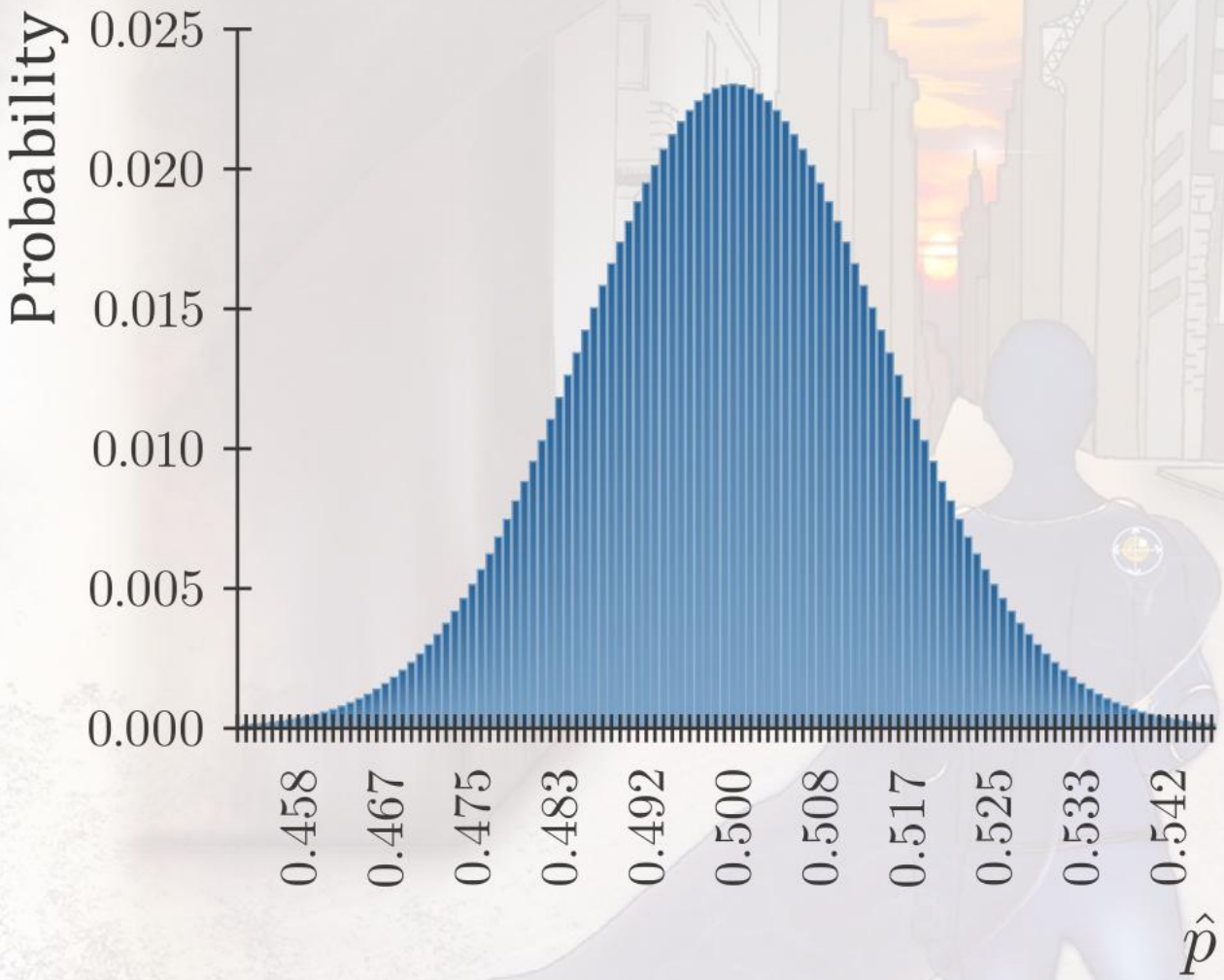
- In theory, the sample proportion can take on a great many values…
  - If $x = 0, \hat{p} = \frac{0}{1200} = 0$
  - If $x = 1, \hat{p} = \frac{1}{1200} = 0.00083$
  - If $x = 2, \hat{p} = \frac{2}{1200} = 0.00167$
  - Etc.
- But hang on: isn't this just a special case of the <u>binomial</u> distribution? ☺
  - $\hat{p}$ is a **random variable**
  - It concerns a binary choice: does a hero play it safe, or take a big financial risk?
  - The number of heroes that get properly insured (x) is a binomial variable. $\hat{p}$ is just x divided by n (a constant)
- For now, let's imagine that the actual population proportion (p) was much better. Suppose that $p = 0.50$. What would that imply for the binomial model?

Sampling distribution



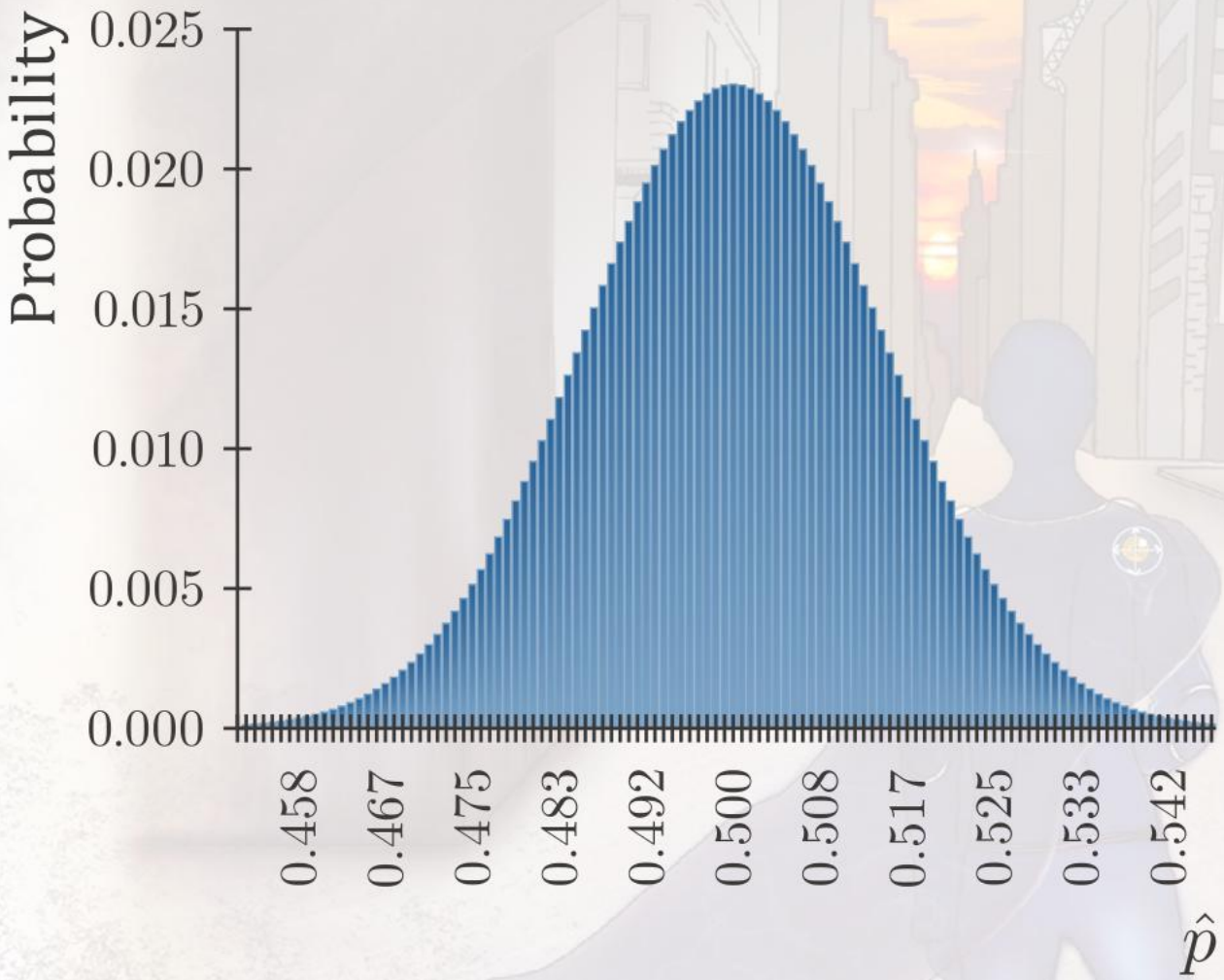
- This distribution describes all the values that $\hat{p}$ could take on, and how likely each value is
- We call it the **sampling distribution of $\hat{p}$**
  - Because it shows us how this statistic may vary from sample to sample
- As you can see, it's pretty much a <u>normal</u> distribution
  - The Success/Failure condition applies:
  - $np = 1200 * 0.50 = 600 > 10$
  - $nq = 1200 * 0.50 = 600 > 10$

A normal distribution!
That allows for useful calculations ☺



- For instance, how about the percentage rule (68-95-99.7%)?
- First, we need to establish the mean and standard deviation of the distribution
- Mean (expected value):
  - For a binomial variable, $\mu = np$
  - A proportion is just x divided by n, so the expected value of $\hat{p}$ becomes
    $$\mu = \frac{np}{n} = \boldsymbol{p}$$
  - OMG – of course
- Standard deviation:
  - For a binomial variable, $\sigma = \sqrt{npq}$
  - Divide this by n, and the standard deviation of $\hat{p}$ becomes
    $$\sigma = \frac{\sqrt{npq}}{n} = \sqrt{\frac{\boldsymbol{pq}}{\boldsymbol{n}}}$$

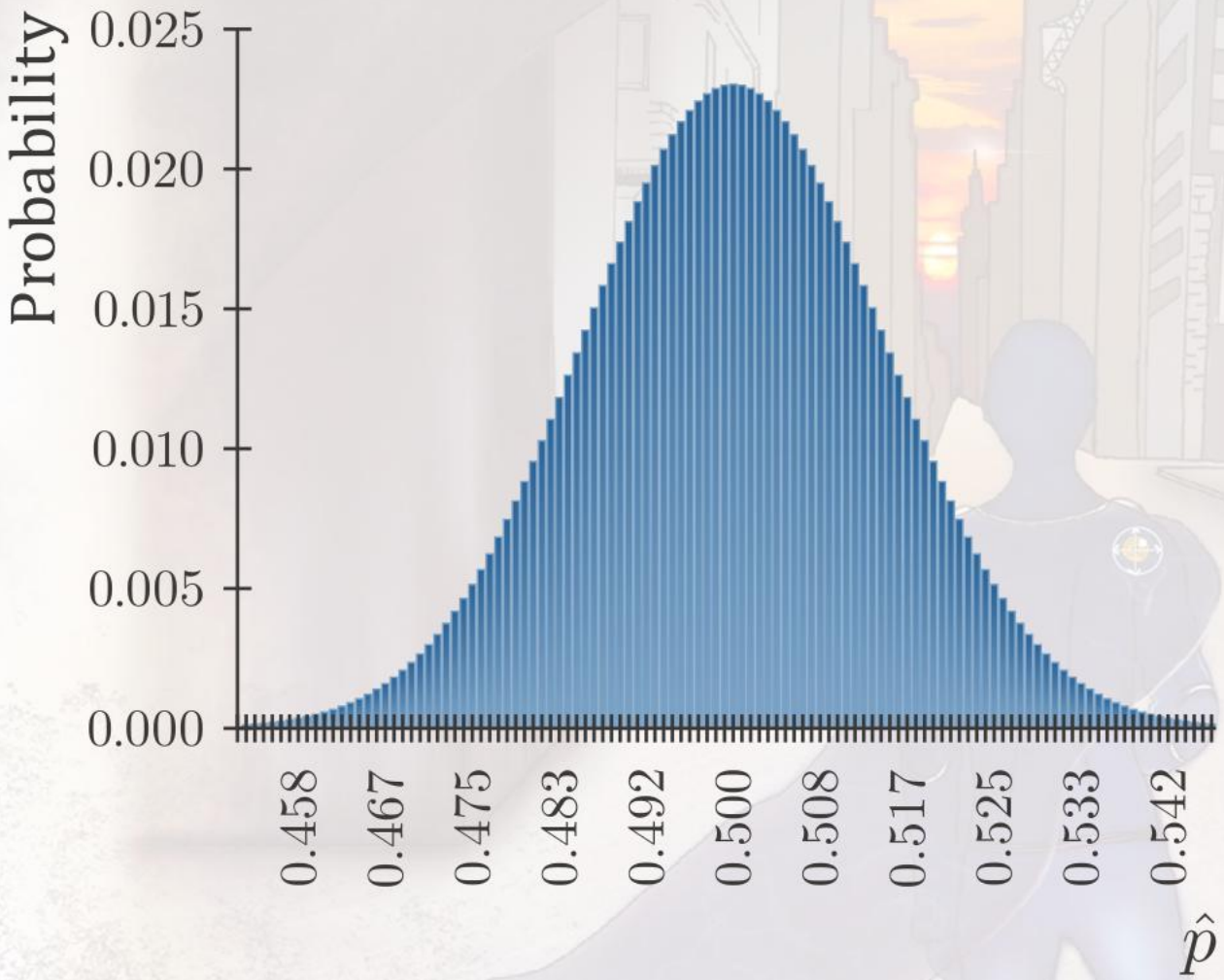
Wrapping up the parameters



- If we assume that 50% of the entire population took precautions for the day they'd break six ribs ($p = 0.50$), then
  - $\mu = 0.50$
    - So: the <u>average</u> sample would get the population proportion exactly right
  - $\sigma = \sqrt{\frac{0.50*0.50}{1200}} = 0.014$
    - So: on average, the sample proportion will deviate from the true *p* by 0.014 (a.k.a. 1.4%)

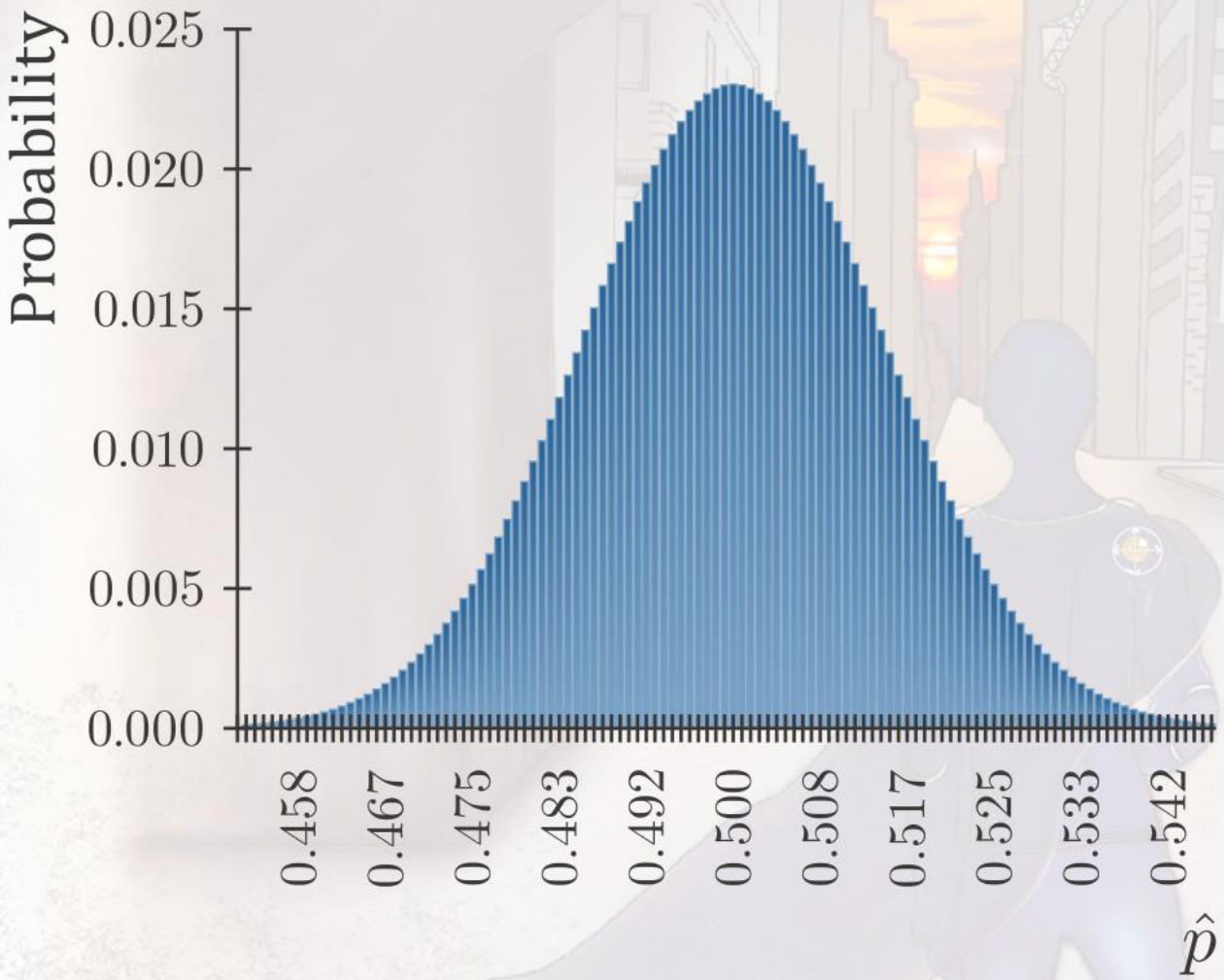
Now let's use the percentage rule

$(\mu = 0.50, \sigma = 0.014)$



- Remember what it says:
  - 68% of the values in a normal distribution will fall within <u>1 standard deviation</u> from the mean
  - 95% will fall within <u>2 standard deviations</u>
  - 99.7% will fall within <u>3 standard deviations</u>
- So if we assume that $p = 0.50$….
  - 68% of the samples (of *n* = 1200) Xavier *could hypothetically draw* would have a sample proportion between
    $$0.50 \pm 0.014 = [0.486, 0.514]$$
  - 95% of the samples Xavier could draw would have a sample proportion between
    $$0.50 \pm 2 * 0.014 = [0.472, 0.528]$$
  - And 99.7% of the times someone drew a sample of this size, the sample proportion would end up between
    $$0.50 \pm 3 * 0.014 = [0.458, 0.542]$$

Reflection



- The study actually found a sample proportion of just $\hat{p} = 0.16$ heroes with supplementary health insurance
- But if the true population proportion equals $p = 0.50$, such a finding would be extremely unlikely
- What should we do with that?
  - Did the researcher just get quite unlucky by stumbling upon such a bizarre, unrepresentative sample?
  - Or was p simply never 0.50 to begin with (but far lower)?
- Maybe we can invent a more elegant method to say something about the population proportion ☺

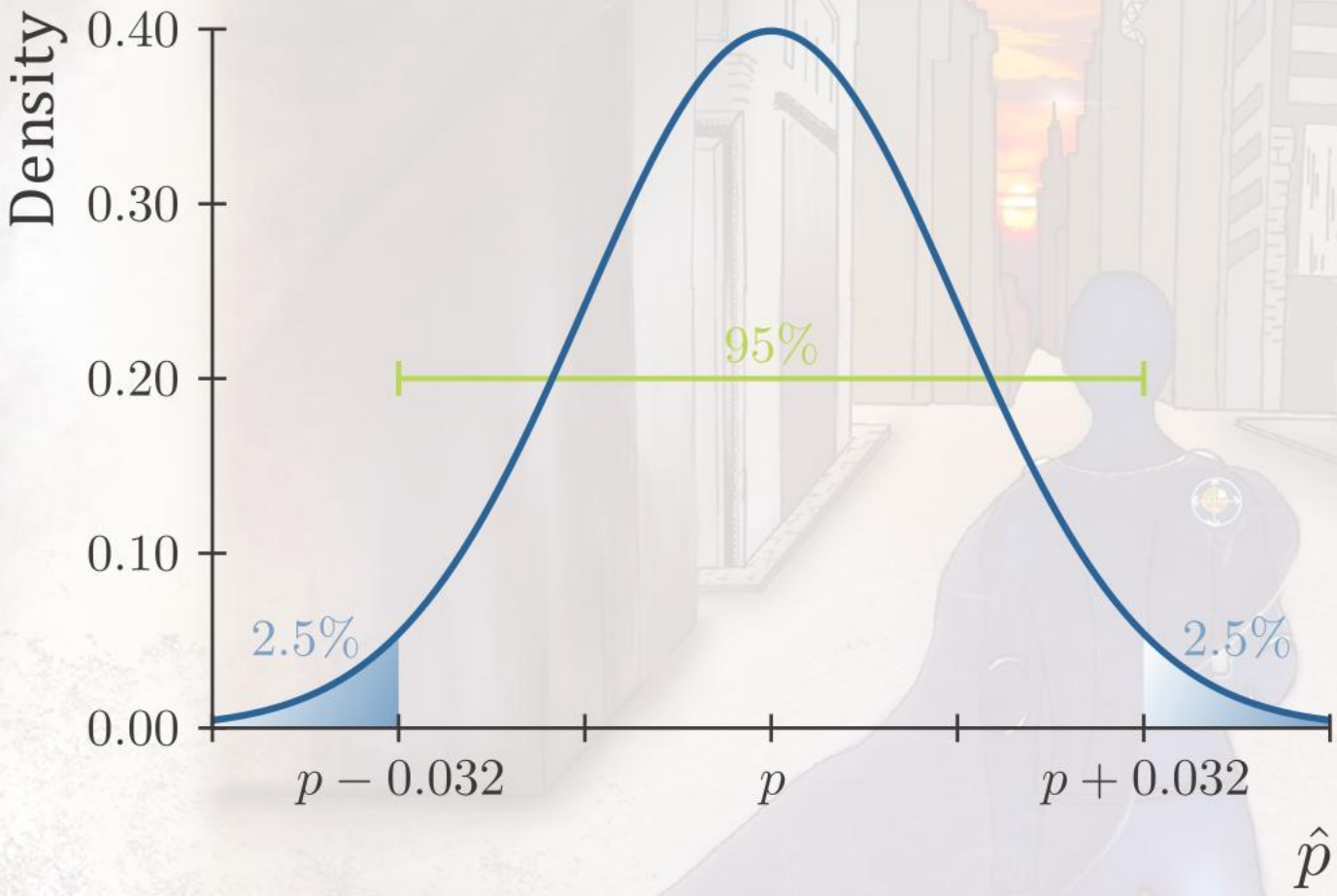
The final boss

Confidence intervals

Rumour has it that many superheroes use performance enhancement drugs

- As part of his survey, Professor Xavier also asked his respondents whether they had taken drugs before at least one mission in the last 12 months
- *n* = 900 heroes answered, and 360 said yes
- The sample proportion is therefore $\hat{p} = \frac{360}{900} = 0.40$
- But what could be the true population proportion?

Imagine the actual sampling distribution



- At its centre lies the true population proportion p, which we don't know
- Where is our $\hat{p}$ on this spectrum? Good question. However…
- This sampling distribution is approximately normal, since the sample is large ☺
- 68% of the time you draw a sample, $\hat{p}$ will fall within *1 standard deviation* from p. How much is one standard deviation?

$$\sigma = \sqrt{\frac{pq}{n}}$$

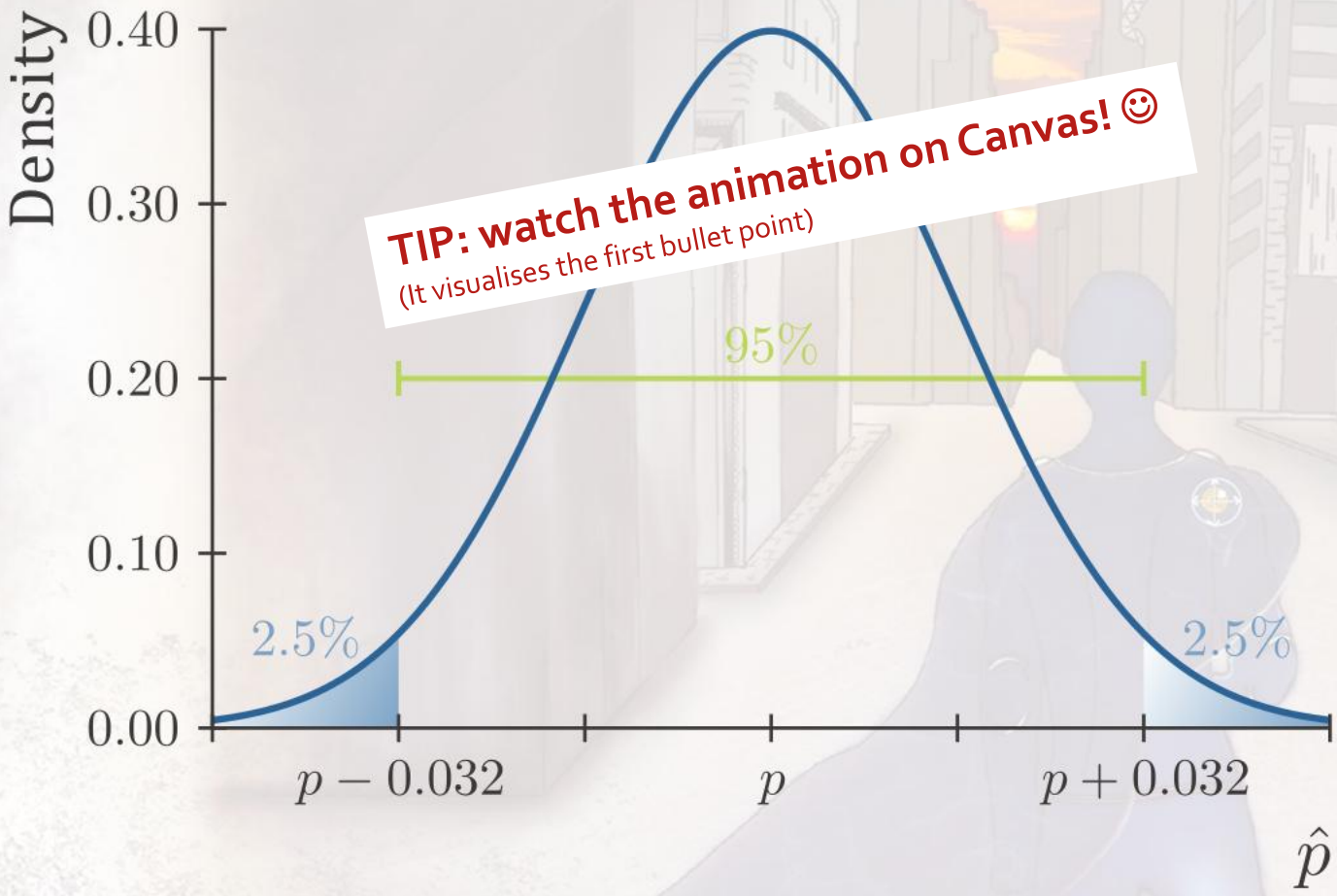
- Sadly, we don't know p or q. But perhaps we can try to <u>estimate</u> σ. This estimator is called the **standard error**:

$$\boldsymbol{SE(\hat{p})} = \sqrt{\frac{\boldsymbol{\hat{p}\hat{q}}}{\boldsymbol{n}}} = \sqrt{\frac{0.40 * 0.60}{900}} = 0.016$$

- Nice. That's 68%. And 95% of the time, $\hat{p}$ will be no more than *2 standard errors* away from p:

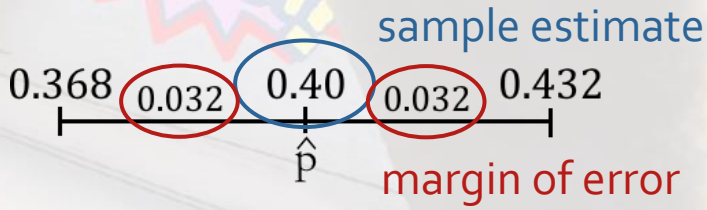
$$2 * 0.016 = 0.032$$

A confidence interval for p



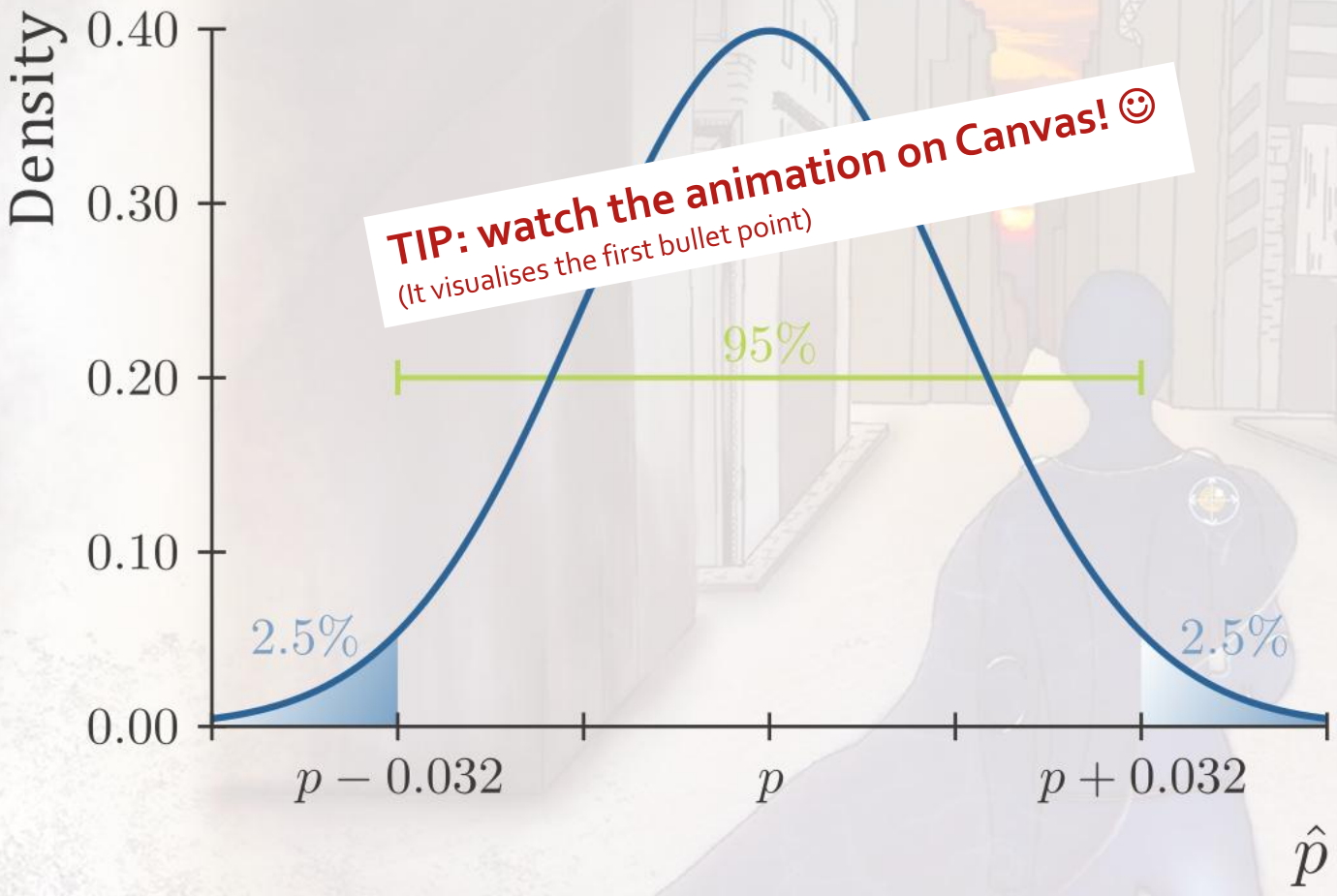
TIP: watch the animation on Canvas! ☺
(It visualises the first bullet point)

- If we drew many samples, 95% of the sample proportions ($\hat{p}$) would be at most 0.032 away from p (that's 3.2 percentage points of heroes that take drugs)
- Reverse the logic: now that we have a $\hat{p}$ (0.40), it's likely that p is no more than 0.032 away!



- This interval,
$$[0.368, 0.432]$$
is called a **95% confidence interval** for the population proportion

A confidence interval for p (continued)



TIP: watch the animation on Canvas! ☺
(It visualises the first bullet point)

- And indeed: for 95% of all samples you could obtain, p would lie inside the 95% confidence interval
- Thus, we can be *95% confident* that p is somewhere inside our current interval:
  $$[0.368, 0.432]$$
- 'I want to be more confident!' (Say, 99%)
- Let's look at the general formula for the confidence interval:

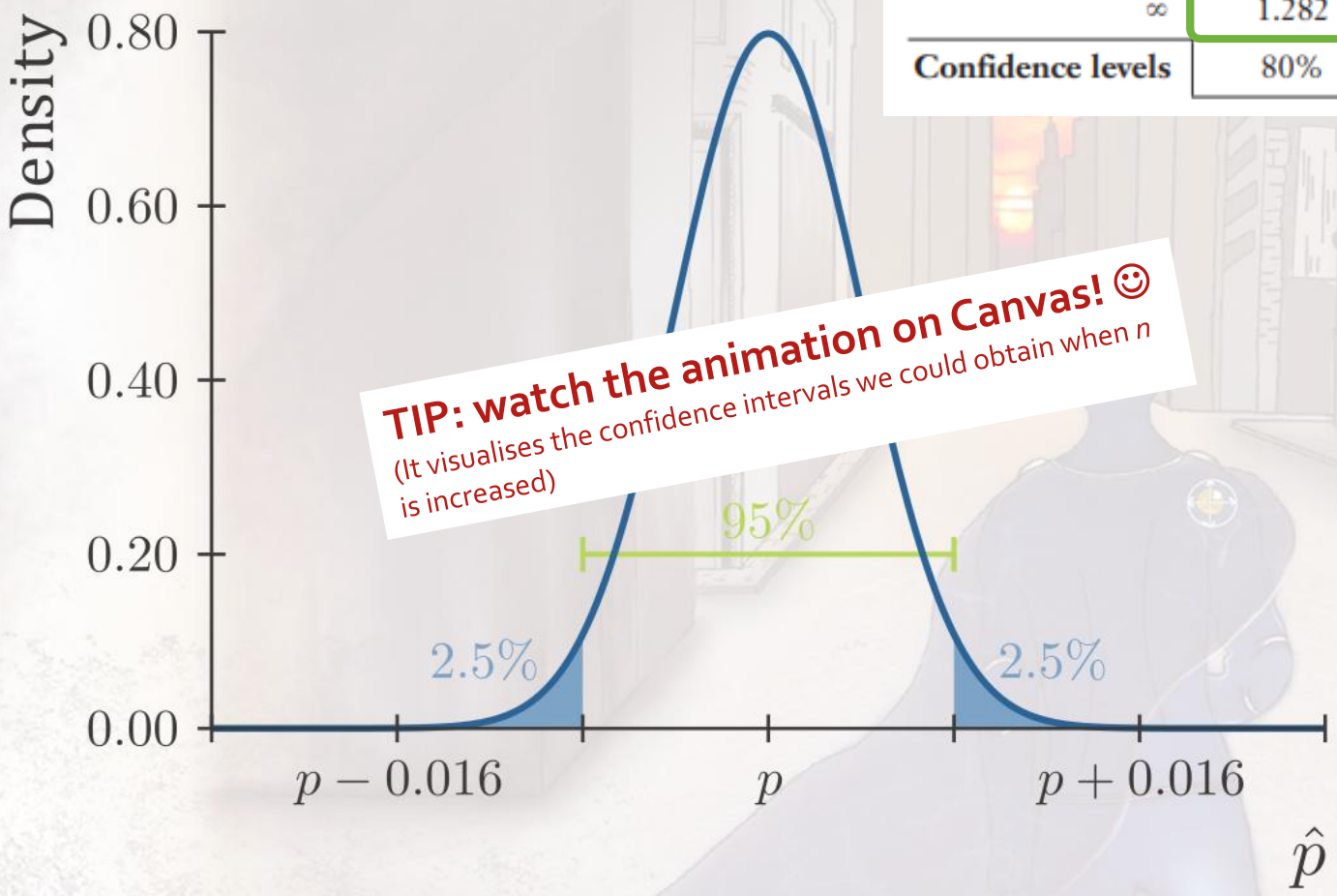
sample estimate

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

margin of error

- consists of a certain number of standard errors
- z^*, the **critical z-value**, indicates *how many standard errors*
- for a 95% interval, $z^* \approx 2$

A better margin of error



TIP: watch the animation on Canvas! ☺
(It visualises the confidence intervals we could obtain when n is increased)

∞	1.282	1.645	1.960	2.326	2.576	z^*	∞
Confidence levels	80%	90%	95%	98%	99%		

bottom of table T in Sharpe

- z^* therefore determines the confidence level
- Increase z^* for greater confidence
- Result: broader interval ☹
  (you can say with less precision what the value of the population proportion is)
- Can't we *narrow down* the interval without hurting the confidence level?
- Oh yes: increase the *sample size*!

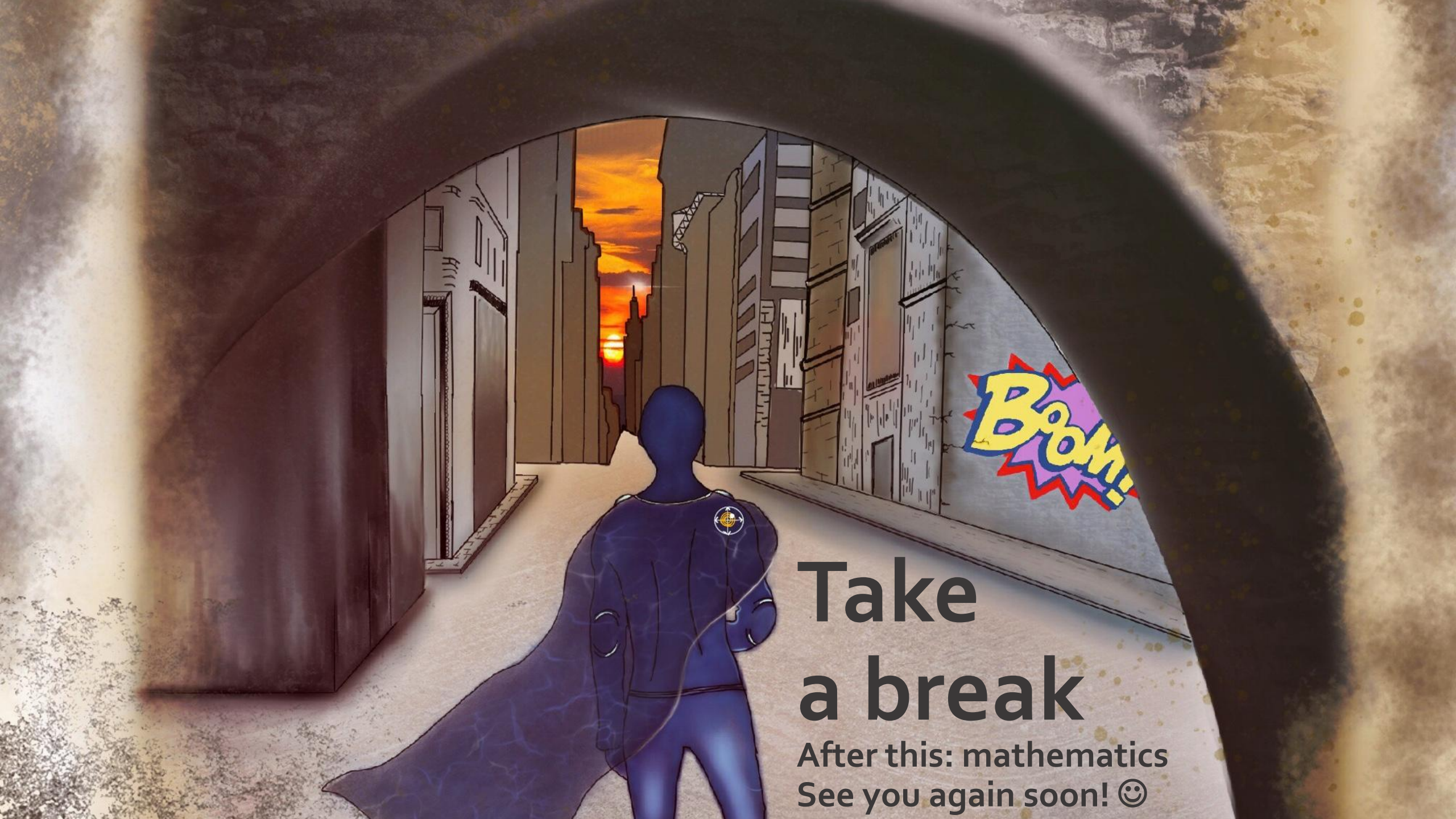
$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

  - A smaller standard error makes the entire distribution narrower, and confidence intervals follow along

Further topics for this week (self-study ☺)

- Sharpe, chapter 9
  - Experiments and the four principles of experimental design
  - Issues in experimental designs
  - Multi-factor designs and interaction
- Sharpe, chapter 10
  - Choosing the sample size (section 10.4)

- Keep in mind: we expect you to read the whole book chapter. This slide highlights the most important aspects that were not covered in this lecture



Take a break

After this: mathematics
See you again soon! ☺